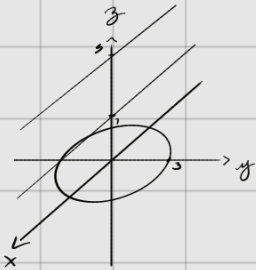


$$\textcircled{1} \int_0^1 \int_1^2 \int_1^3 x + \frac{y}{z} dy dz dx = \int_0^1 \int_1^2 2x + \frac{2}{3}y dy dx = \int_0^1 2x + 1 dx = x^2 \Big|_0^1 + x \Big|_0^1 = 1 + 1 = 2 //$$

$$\int_1^3 x + \frac{y}{z} dy = \int_1^3 x + y z^{-2} dy = 2x - y z^{-1} \Big|_1^3 = 2x - y \left(\frac{1}{3} - 1 \right) = 2x + y \frac{2}{3}$$

$$\int_1^2 2x + \frac{2}{3}y dy = 2x + \frac{2}{3}y \Big|_1^2 = 2x + 1$$

$$\textcircled{2} \iiint y \text{ limitado por } z=1; z=5; y=0; y=\sqrt{9-x^2}$$



$$y^2 + x^2 = 9$$

$$x=3$$

$$y=r \sin \theta$$

$$\begin{cases} 0 \leq x \leq 3 \\ 0 \leq \theta \leq \pi \\ 1 \leq z \leq 5 \end{cases}$$

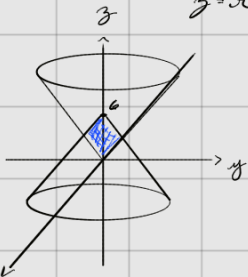
$$\int_1^5 \int_0^\pi \int_0^3 r \sin \theta \cdot r dr d\theta dz = \int_1^5 \int_0^\pi 9 \sin \theta d\theta dz = \int_1^5 18 dz = 72 //$$

$$\int_0^3 r \sin \theta \cdot r dr = \int_0^3 r^2 \sin \theta dr = \sin \theta \cdot \frac{r^3}{3} \Big|_0^3 = \sin \theta \cdot 9 \quad \left| \int_0^\pi 9 \sin \theta d\theta = 9 \cdot (-\cos \theta) \Big|_0^\pi = 9 \cdot (-\cos \pi + \cos 0) = 9 \cdot (1 + 1) = 18 \right.$$

$$\textcircled{3} \text{ Calcular o volume por } z=\sqrt{x^2+y^2} \text{ e } z=6-\sqrt{x^2+y^2}$$

$$z=r$$

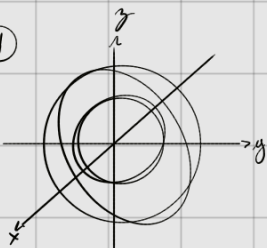
$$z=6-r$$



$$\begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq r \leq 3 \\ r \leq z \leq 6-r \end{cases} dy dz dx$$

$$\int_0^{2\pi} \int_0^3 \int_r^{6-r} r dz dr d\theta = \int_0^{2\pi} \int_0^3 6r - 2r^2 dr d\theta = \int_0^{2\pi} 9 d\theta = 18\pi //$$

④

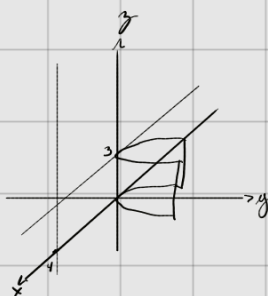


$$\begin{cases} 1 \leq \rho \leq 2 \\ 0 \leq \theta \leq 2\pi \\ 0 \leq \phi \leq \pi \end{cases}$$

$$\int_0^\pi \int_0^{2\pi} \int_1^2 \frac{1}{\rho^2} \rho^2 d\rho d\theta d\phi$$

$$d\rho d\theta d\phi = \int_0^\pi \int_0^{2\pi} 1 d\theta d\phi = 4\pi //$$

$$\textcircled{5} \iiint \frac{1}{\sqrt{x^2+y^2}} \text{ limitado por } z=y^2; z=3; x=0; x=4 \text{ e } y=0$$



$$0 \leq x \leq 4$$

$$0 \leq z \leq 3$$

$$\int_0^{\ln 3} \int_0^{\ln 3} e^{2x-y} dx dy = \int_0^{\ln 3} \int_0^{\ln 3} e^{2x} \cdot e^{-y} dx dy = \int_0^{\ln 3} e^{2x} dx \cdot \int_0^{\ln 3} e^{-y} dy =$$

$$\left(\frac{1}{2} e^{2x} \right) \Big|_0^{\ln 3} \cdot \left(-e^{-y} \right) \Big|_0^{\ln 3} = \left(\frac{1}{2} e^{\ln 2 \cdot 3} - \frac{1}{2} e^0 \right) \cdot \left(e^{\frac{3}{2}} - \frac{1}{2} \right) = \left(\frac{25}{2} - \frac{1}{2} \right) \cdot \left(\frac{1}{3} - \frac{1}{2} \right)$$

$$12 -$$

$$\frac{2}{3 \cdot 2}$$